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B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

SE 19-206-0304 FIRE ENGINEERING FUNDAMENTALS (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Explain the theory of fire chemistry and the different kinds of combustion and their characteristics.
 CO2: Summarize the characteristics of diffusion and premixed flame.
 CO3: Explain the theory of heat transfer mechanism in compartment fires.
 CO4: Describe the use and operation of fire service equipment, machineries and accessories.
 CO5: Calculate the water requirement and the pump capacity for fire fighting.
 CO6: Recognize the basic fire ground operations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Discuss the concept of flammability range with an example.	3	L2	1	1
(b) Determine the equivalence ratio of combustion if 25% excess air is used in a methane gas combustor.	3	L2	1	1
(c) Calculate fire load in a room with size 50 m × 15 m and contains following combustibles: wooden furniture: 200 kg, P U foam furniture: 50 kg. Take heating value of wood as 12 MJ/kg, PU : 25 MJ/kg.	3	L2	3	1
(d) Explain the reason behind the tendency of flames to form a spherical shape in micro gravity conditions (gravity negligible).	3	L3	2	2
(e) Compare dry hydrants and wet hydrants.	3	L2	4	1
(f) Describe any one knot used for rescue operation.	3	L2	4	1
(g) Contrast between Total Energy Line (TEL) and Hydraulic Grade Line (HGL).	3	L1	5	2
(h) Briefly discuss the practical method to estimate the water requirement to tackle building fire.	3	L2	5	1

PART B

(4 × 12 = 48)

II. (a) Give the classification of fire based on fuel as per IS code. Discuss the problems if the recommended extinguishing medium is not used.	6	L2	4	1
(b) Discuss the different factors contributing to a flash fire and dust explosions. Use examples to substantiate the role of each contributing factor.	6	L2	1	1

OR

III. (a) Differentiate between the temperature-time plots of a fuel controlled and ventilation controlled (well ventilated and under ventilated) enclosure fire.	6	L2	3	1
(b) Considering pool fire as an example, discuss the features of diffusion flame. What do you mean 'air entrainment' to such a flame?	6	L2	2	1

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Describe the relevance of convection in the context of an enclosure fire. Explain Newton's law of cooling in convection.	6	L3	2	1,5
	(b) Elucidate the importance of view factor calculations in fire safety. Use the concept of view factor to determine the thermal radiation for the case given: A 500 kW fire develops on a combustible wooden floor. Assuming the combustible floor as the base of an isosceles triangle, find the thermal radiation incident on the other two sides of the triangle. Take radiant heat fraction as 0.3.	6	L3	3	1,5
OR					
V.	(a) LPG gas leaking from a pipe at 0.2 m/s velocity has met with an ignition source. Leak hole diameter is 3 mm. Estimate the flame length using any suitable empirical correlation. [Given: room air temperature: 30°C, air density: 1.2 kg/m ³ , specific heat capacity of air: 1 kJ/kgK, calorific value: 55 MJ/kg]	6	L2	2	1,5
	(b) Define the following terms and give a brief explanation: (i) axi-symmetric fire plume (ii) ceiling jet (iii) response time index.	6	L3	3	2,5
VI.	Explain different fire hoses. Discuss the material of construction, applications and maintenance of each in detail.	12	L1	4	1
OR					
VII.	(a) With neat sketches explain the different ladders used in fire service.	6	L1	4	1
	(b) Compare crash tenders with other fire tenders in terms of their firefighting capabilities and specific features.	6	L1	4	1
VIII.	(a) Discuss nozzle reaction and its management in fire service. How can we estimate the nozzle reaction of different nozzles?	6	L2	5	1
	(b) Elaborate the following: (i) minor losses in pipe network (ii) effect of water hammer and cavitation.	6	L3	5	3,5
OR					
IX.	(a) Explain the relevance and procedure of ventilation operation for a compartment fire.	6	L1	6	1
	(b) Discuss the importance of calculation of pressure drop due to pipe friction in fire service. Calculate the pressure drop for the following situation: water flow rate : 700 litres per minute working pressure of water pump : 7 bar length of the pipe : 100 m Diameter of the pipe : 70 mm Friction factor : 0.005	6	L3	5	3,5

Blooms's Taxonomy Levels

L1 - 27.5%, L2 - 45%, L3 - 27.5%.
